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Full-Information and Information-Constrained Optimal Policies in Reinforcement Learning

Table of Contents

Introduction and classical notions of optimality in reinforcement learning and decision making

Differences between full-information optimal policies and information-constrained optimal policies

Normative formulations of information constraints and reward-complexity trade-offs

Comparing learned behavior to normative benchmarks under limited information, finite samples, or stale information

Introduction and classical notions of optimality in reinforcement learning and decision making

In the classical view, an optimal policy is one that picks the best action in each state to maximize long-run return, often assuming the environment is known or can be learned well enough. In practice, RL usually works under unknown dynamics, partial observation, limited samples, or limited compute, so "optimality" often shifts from exact statewise optimal control to the best behavior attainable within those limits. (6 sources)

Classical decision making in Markov decision processes starts from a strong benchmark: the environment is represented by states, actions, rewards or costs, and transition dynamics, and the goal is to compute a policy that is optimal in the dynamic-programming sense—best action choice in every state with future consequences taken into account. This offline MDP view assumes the model is known in advance and supports forward-looking optimization by dynamic programming. (Guan et al., 2014) (Felten et al., 2022) In that classical picture, reinforcement learning keeps the same target policy concept, but drops the assumption of full knowledge: instead of being given the dynamics and rewards, the agent must learn by interacting with the environment. (Felten et al., 2022) (Guan et al., 2014)

A central classical idea is the principle of optimality: an optimal policy should act optimally in every state that can occur, not just do well from the initial condition on average. Thomas et al. note that this statewise principle sits at the foundation of classical RL, and that methods such as Q-learning reflect it by trying to choose optimally in each encountered state. They also point out that a start-state objective can differ from this stronger notion, especially with function approximation and limited state coverage. (Thomas et al., 2019) This distinction matters because "optimal" can mean either globally best expected return from a start distribution or locally best continuation from each state, and the two can come apart once approximation enters. (Thomas et al., 2019)

The classical benchmark is also easiest under full information. With perfect information about the environment, one can often describe an optimal policy as a mapping from observation or state to action and compute it efficiently relative to the model size and horizon. But once the state is only partly observed, or the model is unknown, the policy may need to depend on the whole interaction history, and exact optimization becomes much harder. In realistic settings, one therefore searches for near-optimal policies within a restricted policy class rather than expecting to find the absolute best policy. (Peshkin et al., 2001) This is one reason classical optimality is best seen as a normative ideal, while practical RL often aims for approximate solutions.

Another classical notion is the optimal value function: the long-run cumulative reward achievable from a

state, or state-action pair, when optimal choices are made thereafter. In RL, this object guides learning because the agent can estimate values from sampled experience and then use those estimates to choose actions. But these estimates are built from limited samples rather than full enumeration of all state-action pairs, so even “optimal” behavior is usually with respect to what has been sampled and learned so far. (Kim et al., 2023) This sample-based character is one of the main ways RL differs from exact planning.

The contrast between planning and learning is especially clear when future structure must be inferred rather than directly observed. Schmidhuber’s box example shows that a policy that is myopically best given past outcomes can still be worse than one that learns the hidden regularity and predicts future rewards. The broader lesson is that RL is not just search over immediate empirical payoffs; it is about using experience to improve future sequential decisions under uncertainty. At the same time, ideal universal notions of optimal RL are often not feasible in practice, so the real target is good use of limited computation and data. (Schmidhuber, 2002)

Finally, some sequential decision settings change the benchmark altogether. Standard RL usually assumes a stochastically stable environment so that an optimal stationary policy exists and can in principle be learned. When costs vary arbitrarily over time, the right benchmark may no longer be an optimal stationary policy but low regret relative to a hindsight comparator. This shows that “optimality” in decision making is not one fixed concept; it depends on what the agent knows, what stays stable, and what comparator class is reasonable. (Guan et al., 2014)

Evidence

(Guan et al., 2014)

Online Markov Decision Processes With Kullback–Leibler Control Cost

> *"In the basic MDP framework, it is assumed that the cost functions and the transition probabilities are known in advance, the policy is designed "offline" (e.g., using dynamic programming), and the optimality criterion is forward-looking, taking into account the effect of past actions on future costs. In many practical problems, however, this degree of advance knowledge is unavailable. When neither the transition probability nor the cost functions are known in advance, various reinforcement learning (RL) methods, such as the celebrated Q-learning algorithm (Watkins et al., 1992), (Tsitsiklis, 1994) and its variants, can be used to learn an optimal policy in an online regime. However, the key assumptions underlying RL are that the agent is operating in a stochastically stable environment, and that the state-action costs (or at least their expected values with respect to any environmental randomness) do not vary with time. These assumptions are needed to ensure that the agent is eventually able to learn an optimal stationary control policy"*
>
> *". In this online learning (or sequential prediction) framework, the effects of the environment are modeled by an arbitrarily varying sequence of cost functions, where the cost function at each time step is revealed to the agent only after an action has been taken. There is no state, and the goal of the agent is to minimize regret, i.e., the difference between the total cost incurred using causally available information and the total cost of the best single action that could have been chosen in hindsight"*
>
> *". This paper considers an online (real-time) control problem"*
>
> *"An explicit construction of a computationally efficient strategy with small regret (i.e., expected difference between its actual total cost and the smallest cost attainable using noncausal knowledge of the state costs) under mild regularity conditions is presented"*

(Felten et al., 2022)

Metaheuristics-based Exploration Strategies for Multi-Objective Reinforcement Learning

> *"The main difference between RL and dynamic programming (DP) is that the latter assumes a perfect knowledge of the environment dynamics. Indeed, in DP, the (S, A, r) tuple is known in advance and the algorithm searches for the optimal policy whereas RL does not know anything about the environment and learns π * by interacting with it."*

(Thomas et al., 2019)

Classical Policy Gradient: Preserving Bellman's Principle of Optimality

> *"This principle of optimality has endured at the foundation of reinforcement learning research, and is central to what remains the classical definition of an optimal policy [2]. Classical reinforcement learning algorithms like Q-learning [3] embody this principle by striving to act optimally in every state that occurs, regardless of when the state occurs"*
>
> *".The start-state objective function, ρ(θ) = E[∑ ∞ t=0 γ t R t |θ], prioritizes making decisions optimally in the initial state, not*

necessarily in the states resulting from the first decisions"

".These two goals (optimizing decisions in the initial state and optimizing decisions in subsequent states) can be conflicting when using function approximation, particularly when γ is small and the initial state distribution has limited support. So, maximizing ρ does not preserve the principle of optimality"

".This form harks back to the classical definition of an optimal policy, particularly if d θ has full support on S and does not depend on θ, in which case f preserves the partial ordering on policies used in the classical definition of an optimal policy"

".So, we trade-off similarity to the classical definition of an optimal policy with the practicality of estimating the objective function and its gradient"

(Peshkin et al., 2001)

Bounds on Sample Size for Policy Evaluation in Markov Environments

"There are many efficient algorithms for the case when the agent has perfect information about the environment. An optimal policy is described by mapping the last observation into an action and can be computed in polynomial time in the size of the state and action spaces and the effective time horizon [2]"

".However in many cases the environment state is described by a vector of several variables, which makes the environment state size exponential in the number of variables. Also, under more realistic assumptions, when a model of environment dynamics is unknown and the environment's state is not observable, many problems arise. The optimal policy could potentially depend on the whole history of interactions and for the undiscounted finite horizon case computing it is pspace-complete (Papadimitriou et al., 1987). In realistic settings, the class of policies is restricted and even among the restricted set of policies, the absolute best policy is not expected to be found due to the difficulty of solving a global multi-variate optimization problem. Rather, the only option is to explore different approaches to finding near-optimal solutions among local optima"

".The issue of finding a near-optimal policy from a given class of policies is analogous to a similar issue in supervised learning. There we are looking for a near-optimal hypothesis from a given class of hypotheses [28]"

".On the other hand, there is no fixed distribution generating experiences in reinforcement learning. Each policy induces a different distribution over experiences. The choice of a policy defines both a "hypothesis" and a distribution. This raises the question of how one re-uses the experience obtained while following one policy to learn about another. The other policy might generate a very different set of samples (experiences), and in the extreme case the support of the two distributions might be disjoint."

(Kim et al., 2023)

Deep Q-Network Approach for Train Timetable Rescheduling Based on Alternative Graph

"Reinforcement learning is one way to mitigate the computational limitations of an enumerative search of the solution space, by relaxing the goal of finding any exact optimal solutions. To do this, instead of searching all possible pairs of states and actions, a relatively small samples of states and actions is simulated and an optimal or near-optimal solution for the problem with respect to these samples is deduced. One of the central concepts in reinforcement learning is the optimal value function of a state, or pair of states, and an action, which specifies the long-term cumulative reward achievable when the agent follows (possibly unknown) an optimal decision after the given state or the given pair of states and action. For a given sample, we can estimate the optimal value function, and the estimated value function can be used directly or indirectly to specify the decisions the agent makes."

(Schmidhuber, 2002)

Optimal Ordered Problem Solver

"Sometimes, however, it won't. To see the difference between searching (the topic of the previous sections) and reinforcement learning (RL), consider an agent and two boxes. In the n-th trial the agent may open and collect the content of exactly one box. The left box will contain 100n Swiss Francs, the right box 2 n Swiss Francs, but the agent does not know this in advance. During the first 9 trials the optimal policy is "open left box." This is what a good searcher should find, given the outcomes of the first 9 trials. But this policy will be suboptimal in trial 10. A good reinforcement learner, however, should extract the underlying regularity in the reward generation process and predict the future reward, picking the right box in trial 10, without having seen it yet"

".However, while AIXI clarifies the theoretical limits of RL, it is not practically feasible, just like Hsearch is not. From a pragmatic point of view, what we are really interested in is a reinforcement learner that makes optimal use of given, limited computational resources."

Differences between full-information optimal policies and information-constrained optimal policies

Full-information optimal policies assume the agent can use all relevant state information and enough computation to

maximize return exactly. Information-constrained optimal policies change the benchmark: the agent should get as much reward as possible given limits on observation, memory, computation, or policy complexity. (11 sources)

A full-information optimal policy is the classical ideal: if the agent can observe the true state, then in an MDP the optimal policy can be memoryless, choosing actions from the current state alone to maximize expected return. But this benchmark quietly assumes both full observability and enough computational power to search for exact optimal actions. Several recent papers make this point directly: standard MDP-based RL usually assumes full state observability, which implies a memoryless optimal policy, while partial observation breaks that property and can force the optimal policy to depend on the whole history instead (Zhang et al., 2023) (Vitale et al., 2025) (Jordan et al., 2026). In a POMDP, the agent only gets partial or noisy observations of latent state, so acting optimally may require remembering past observations and actions; this makes the effective decision problem much larger and often exponentially harder in the horizon (Zhang et al., 2023) (Jordan et al., 2026) (Meuleau et al., 1999).

Information-constrained optimality starts from a different premise: the agent is still trying to maximize reward, but only within a fixed information-processing budget. In this view, the policy itself is treated as an information channel from states or observations to actions, and optimality is defined by a constrained optimization problem that trades reward against the information needed to implement the policy (Malloy et al., 2020). Larsson et al. contrast this with the traditional MDP view, noting that classical frameworks ignore computational limits and effectively assume perfect rationality, whereas bounded-rational models define the best policy as the one that gives the highest expected value for a given processing cost (Larsson et al., 2017). In this sense, an information-constrained policy may be “suboptimal” under the unconstrained benchmark yet fully optimal under the right constrained benchmark.

A common normative form of this constrained benchmark is to maximize expected utility subject to a cap on information capacity, or equivalently to maximize utility minus an information cost. Malloy et al. write this explicitly as reward maximization with a mutual-information constraint on the policy channel (Malloy et al., 2020). Closely related free-energy formulations measure the cost of moving from a default or prior policy to a more tailored posterior policy using relative entropy, so the agent is penalized for making highly state-specific choices unless the reward gain justifies that cost (Pushp et al., 2023) (Ortega et al., 2012). This makes the difference from classical optimality very clear: full-information optimal control asks “what action maximizes return?”, while bounded-rational control asks “what action rule gives the best return improvement per bit or per unit of computation?” (Larsson et al., 2017) (Ortega et al., 2012).

These constraints can reflect different limits. They can stand for limited observation of the world, as in POMDPs; limited memory, which prevents conditioning on full history; limited compute, which restricts how many trajectories or plans can be evaluated; or limited policy complexity, which keeps behavior near a simple default policy (Zhang et al., 2023) (Pushp et al., 2023) (Garrido-P'erez et al., 2026). Under these limits, the optimal policy is often stochastic and compressed rather than fully state-contingent, because randomization or reuse of a simpler action rule can save information-processing cost (Malloy et al., 2020) (Garrido-P'erez et al., 2026). This reward-complexity trade-off is central in resource-rational accounts of behavior: more complex policies can earn more reward, but they also require more memory or information capacity (Garrido-P'erez et al., 2026).

So the key difference is not just that constrained agents perform worse. It is that the objective itself changes. Under full information, the benchmark is the globally return-maximizing policy. Under information constraints, the benchmark is the policy on the relevant Pareto frontier between reward and information cost: no other feasible policy can get more reward with the same or lower information use, or use less information for the same reward (Malloy et al., 2020) (Larsson et al., 2017) (Garrido-P'erez et al., 2026). This is why bounded-rational and capacity-limited Bayesian RL are best seen not as weaker versions of classical optimality, but as different normative theories of what “optimal” means when information processing is scarce (Hihn et al., 2019) (Arumugam et al., 2023).

Evidence

(Zhang et al., 2023)

Provable Representation with Efficient Planning for Partially Observable Reinforcement Learning

"the success of MDP-based RL algorithms (Zhang et al., 2022)(Ren et al., 2022) relies on an assumption that state information is fully observable, which implies that the optimal policy is memoryless, i.e., optimal actions can be selected based purely on the current state (Ren et al., 2021)"

".The Partially Observed Markov Decision Process (POMDP) (Åström, 1965) has been proposed to extend the classical MDP formulation by introducing observation variables that only give partial information about the underlying latent state (Hauskrecht et al., 2000)(Roy et al., 2002)(Chen et al., 2016). This extension greatly expands the practical applicability of POMDPs over MDPs, but the additional uncertainty over the underlying state given only observations creates a non-Markovian dependence between successive observations, even though Markovian dependence is preserved between latent states. Consequently, the optimal policy for a POMDP is no longer memoryless but entire-history dependent, expanding the state complexity exponentially w.r.t. horizon length."

(Vitale et al., 2025)

A Flexible Multi-Agent Deep Reinforcement Learning Framework for Dynamic Routing and Scheduling of Latency-Critical Services

"The literature also analyzes settings where the agent does not observe the full environment state but instead receives a partial observation o = ω(s). This scenario is referred to as a Partially Observed MDP (POMDP) (Kaelbling et al., 1998). In this case, actions are selected based on the partial observation, following a policy π(a|o)."

(Jordan et al., 2026)

Model-Based Learning of Near-Optimal Finite-Window Policies in POMDPs

"MDPs have become the standard model in reinforcement learning, they rely on the agent having full knowledge of the environment's state -an idealization that is often unrealistic in practice. POMDPs relax this requirement by allowing the agent to receive only partial or noisy observations of an underlying state"

"it has long been recognized that the greater generality of POMDPs comes with inherent hardness barriers, arising from the fact that optimal policies may depend on the full history of observations"

"As a result, a large body of work has explored approximate solution methods that trade optimality for computational tractability or sample-efficiency. In this work, we focus on approximate approaches that are implementable in a computationally efficient manner. One such approach is the use of finite-window techniques, where policies depend only on a fixed-length history window of recent actions and observations [6], (Loch et al., 1998), (Meuleau et al., 1999)."

(Meuleau et al., 1999)

Solving POMDPs by Searching the Space of Finite Policies

"Solving partially observable Markov decision processes (POMDPS) is highly intractable in general, at least in part because the optimal policy may be infinitely large. In this paper, we explore the problem of finding the optimal policy from a restricted set of policies, represented as finite state automata of a given size. This problem is also intractable, but we show that the complexity can be greatly reduced when the POMDP andlor policy are further constrained. We demonstrate good empirical results with a branch-and-bound method for finding globally optimal deterministic policies, and a gradient-ascent method for finding locally optimal stochastic policies."

(Malloy et al., 2020)

Deep RL With Information Constrained Policies: Generalization in Continuous Control

"Agents with limited information processing capacities should seek to produce behavior that maximizes expected utility. Hence, for any agent, optimal behavior is the result of solving a constrained optimization problem (maximizing the utility of behavior, subject to constraints on available information capacity)"

".In particular, we consider an agent's policy as an information channel, that maps from the current state of the environment, to a probability distribution over actions, and define optimal behavior subject to a constraint on the information capacity of this channel"

".This allows us to define the optimality condition of an RL agent with a constraint on information representation as max π 0:T E (st,at)□ ρπT t=0 r(s t , a t) s.t E (st,at)□ ρπ[I(π(a|s)"

"The introduction of this policy mutual information term constrains the agent's policy objective in relation to the information capacity."

(Larsson et al., 2017)

Hierarchical state abstractions for decision-making problems with computational constraints

> *"The traditional (MDP) framework ignores computational limitations while searching for optimal policies, essentially assuming that the acting agent is perfectly rational and aims for exact optimality"*
>
> *".The introduction of the added constraints (computational and otherwise) to the decision process is meant to model an agent that does not have access to unlimited resources, and hence will select actions that are not optimal in the traditional sense of maximizing value or utility, but rather in the sense of maximizing (expected) value or utility for a given processing cost (Lipman, 1995)(Tishby et al., 2011)(Rubin et al., 2012)."*

(Pushp et al., 2023)

Coordination of Bounded Rational Drones Through Informed Prior Policy

> *"Consequently, the assumption of perfect rationality, commonly adopted in the standard game-theoretic framework, typically cannot be realized in the real world, as agents are constrained by their computational resources, forcing them only to evaluate a limited and finite number of trajectories to make a decision.*
>
> *Instead, information-theoretic bounded rationality offers a more realistic modeling framework by explicitly considering agents' computational limits to trade-off between the available computational resources and the number of trajectories for evaluation [5,21]27]. This is achieved by defining a constraint considering the amount of information (computation) the agent can afford to transfer from a default (or prior) policy q i to the optimal one. Intuitively, this default policy describes the nominal behavior of the agent before initiating the planning process, and the bounded-rational agent seeks to identify an optimized posterior policy that falls within the neighborhood of the default policy."*

(Ortega et al., 2012)

Thermodynamics as a theory of decision-making with information-processing costs

> *"Perfectly rational decision-makers maximize expected utility, but crucially ignore the resource costs incurred when determining optimal actions. Here, we propose a thermodynamically inspired formalization of bounded rational decision-making where information processing is modelled as state changes in thermodynamic systems that can be quantified by differences in free energy. By optimizing a free energy, bounded rational decision-makers trade off expected utility gains and information-processing costs measured by the relative entropy. As a result, the bounded rational decision-making problem can be rephrased in terms of well-known variational principles from statistical physics. In the limit when computational costs are ignored, the maximum expected utility principle is recovered. We discuss links to existing decision-making frameworks and applications to human decision-making experiments that are at odds with expected utility theory. Since most of the mathematical machinery can be borrowed from statistical physics, the main contribution is to re-interpret the formalism of thermodynamic free-energy differences in terms of bounded rational decision-making and to discuss its relationship to human decision-making experiments."*

(Garrido-P'erez et al., 2026)

Including the Cost of Irreducible Uncertainty in the Policy Compression Framework

> *"Humans have limited cognitive and memory resources"*
>
> *"As a result, decisionmaking is shaped by a reward-complexity trade-off: more complex policies can produce higher rewards, but they also require greater memory resources"*
>
> *"The resource-rational framework proposes that humans solve the aforementioned trade-off optimally [1], meaning that we choose the best possible action subject to our cognitive constraints"*
>
> *"At the same time, it introduces additional complexity for fitting the model to human data, which future work will need to address."*

(Hihn et al., 2019)

An Information-theoretic On-line Learning Principle for Specialization in Hierarchical Decision-Making Systems

> *"Information-theoretic bounded rationality describes utility-optimizing decision-makers whose limited information-processing capabilities are formalized by information constraints. One of the consequences of bounded rationality is that resource-limited decision-makers can join together to solve decision-making problems that are beyond the capabilities of each individual."*

(Arumugam et al., 2023)

Bayesian Reinforcement Learning With Limited Cognitive Load

> *"This body of work provides an account of capacity-limited Bayesian reinforcement learning, a unifying normative framework for modeling the effect of processing constraints on learning and action selection."*

Normative formulations of information constraints and reward-complexity trade-offs

Information-constrained decision models make the trade-off explicit: more tailored, state-specific behavior can earn more reward, but it costs bits, memory, or policy change. The right normative benchmark is therefore not maximum reward alone, but the best reward achievable at a given information or sample budget. (6 sources)

A useful way to make information limits normative is to write decision making as an optimization with two terms: expected reward and an information cost. One common version measures how far the chosen policy moves from a prior, default, or nominal policy using KL divergence or relative entropy. In that setup, the agent is allowed to improve reward, but only by paying for policy specialization; large state-dependent departures from the prior cost more than small ones. Xu et al. describe bounded rationality in exactly this way, with information-processing ability represented by the KL divergence between nominal and optimized stochastic policies, and with the optimized control law taking an exponential-family form closely related to approximate Bayesian inference (Xu et al., 2022). Sledge et al. present the same idea under the "value of information": the agent seeks a transformed version of an initial policy that improves reward, but the total divergence from that initial policy is bounded, so tighter bounds force more conservative changes while looser bounds allow more reward-seeking adaptation (Sledge et al., 2017) (Sledge et al._1, 2017).

This family of objectives can be read in two equivalent ways. In constrained form, the agent maximizes reward subject to an information budget. In regularized form, it maximizes reward minus a penalty for complexity or policy change. Lai et al. note that some formulations define the constraint relative to the full-information optimal policy, while others define it relative to a prior action preference; the latter can be viewed as policy compression because the prior acts like a state-independent code for actions (Lai et al., 2024). This matters because it changes what counts as "simple." A policy is simple not because it gets low reward, but because it reuses the same action tendencies across many states and therefore needs fewer bits to describe.

The reward-complexity trade-off also gives a concrete meaning to coarse versus fine control. When policy complexity is low, many states are effectively grouped together and trigger the same action, so search is coarse and behavior leans toward reuse of known actions. When policy complexity is high, the policy can encode finer distinctions between states, search more broadly, and support more exploratory or risk-tolerant behavior. Sledge et al. make this link explicit, describing policy complexity as the number of bits needed to encode the policy and showing that lower bit budgets induce quantization of the state-action mapping, while higher budgets allow more differentiated action choice (Sledge et al., 2018) (Sledge et al., 2017). In this sense, "complexity" is not just a descriptive label; it is the formal resource that controls how tailored the policy can be to the state.

The same logic applies before action selection, at the level of representation learning. A representation should be informative enough to support good policies, but not so rich that it becomes hard to learn or use with limited-capacity function approximators. Echchahed et al. argue that an optimal representation balances information capacity and simplicity: too much information can slow learning and hinder convergence, while too little prevents convergence to an optimal policy at all (Echchahed et al., 2025). So the reward-complexity trade-off is not only about the final policy channel; it can also appear in the learned state space itself.

Taken together, these formulations say that normative rationality under limits is a frontier, not a single unconstrained optimum. For each allowed level of policy complexity, policy change, coding rate, or representational capacity, there is a best achievable reward. That is why Lai et al. stress that the reward-maximizing policy is no longer the right benchmark once complexity costs matter, and also why comparison to learned behavior must respect the learner's actual information and sample limits rather than an unattainable reward-complexity optimum computed with more information than the learner had (Lai et al., 2024).

Evidence

(Xu et al., 2022)

Decision-Making Among Bounded Rational Agents

"bounded rationality directly models the agent's information processing ability, which is represented as the KL-divergence between nominal and optimized stochastic policies"

"When computing the optimal policy, it includes an additional information-theoretic cost measured by the KL-divergence between a prior policy distribution and a posterior after optimization. This is similar to the effect of the informationtheoretic constraints of bounded rationality, where the final optimal control distribution can be sampled from the exponential family using approximate Bayesian inference [10](Williams et al., 2017)."

(Sledge et al., 2017)

Analysis of Agent Expertise in Ms. Pac-Man Using Value-of-Information-Based Policies

"Stratonovich proposed a mathematical framework for bounded, rational decision making under information constraints. This took the form of an optimization criterion known as the value of information"

". The value of information is based on the premise that there is a trade-off between policy information and obtainable rewards. That is, the greater the amount of information, the better the long-term rewards should be"

". The value of information is a constrained optimization problem that seeks a transformed version of this initial policy that yields the highest rewards. The initial policy is not necessarily allowed to change arbitrarily, though. Rather, the transformation is governed by a information-constraint term that assesses the divergence between the initial policy and the current policy. The total divergence cannot exceed a specified bound. The lower the bound, the less that the initial policy can be modified during learning. This will cause the agent to implement risk-adverse action-selection behaviors, which can impact the obtainable rewards if the initial policy does not encode the environment dynamics well. Conversely, the higher the information bound, the higher the diversity of policies and hence the greater the reward improvement. The agent will therefore favor risky actions."

(Sledge et al._1, 2017)

An Analysis of the Value of Information When Exploring Stochastic, Discrete Multi-Armed Bandits

"Stratonovich proposed a mathematical framework for quantitative decision making under information constraints, which took the form of an optimization criterion known as the value of information. The value of information is based on the premise that the inflow of novel information can be used to increase rewards in decision-making problems. That is, the more environmental details that are made available to the agent, the more informed its decisions will be and hence the better the rewards should be. Here, information will take the form of policy transformation costs"

".The value of information is an optimization problem that seeks a transformed version of this initial policy that yields the highest rewards. The initial policy is not necessarily allowed to change arbitrarily, though, across each arm-pull round. Rather, the transformation is governed by a bounded information constraint term. The lower the bound, the less that the initial policy entries can be modified during a single arm pull. This behavior can impact the obtainable rewards if the initial policy is poor. Conversely, the higher the bound, the more that the initial policy can be adjusted in a given round and hence the greater the potential reward improvement."

(Lai et al., 2024)

Human decision making balances reward maximization and policy compression

"Bounded rationality is the idea that decision-makers try to optimize behavior subject to information-processing constraints [1–[5]"

". In some formulations, these constraints are defined by deviations from the full-information optimal policy [6], while in other formulations they are defined by deviations from some prior action preference [7–[14]. This latter class of models can be interpreted as policy compression [15], since the prior can be viewed as a state-independent code for actions. In this paper, we use the policy compression framework to study how humans trade reward for complexity"

". One implication of this perspective is that the reward-maximizing policy is no longer the appropriate normative benchmark. Instead, the normatively optimal policy should take into account the trade-off between reward and complexity"

". an agent may be unable to estimate the optimal reward-complexity trade-off. Thus learned behavior should not be compared to the reward-complexity optimal policy, but rather to the optimal policy of an agent with the same information or finite sample constraints [53–[55]."

(Sledge et al., 2018)

Guided Policy Exploration for Markov Decision Processes Using an Uncertainty-Based Value-of-Information Criterion

> *"The value of information implements an information-theoretic [6] optimization criterion. It describes the maximum benefit that can be gained by a given quantity of information in order to minimize average losses. When applied to Markov decision processes, the criterion provides an optimal trade-off between policy cost and the granularity of the search process (Sledge et al., 2017). The granularity is dictated by the number of bits needed to encode the policy. When the number of bits is low, the amount of information that the states provide about the actions is artificially limited. The policy space is searched coarsely and the exploitation of previously learned behaviors dominates. This is due to a quantization of the action-state space: many states will be grouped together and visiting them will elicit the same action response. When the number of bits is high, the amount of information carried about the actions by the states is increased. The policy space is thoroughly surveyed and the exploration of new actions dominates. There is little to no quantization of the action-state space in this case."*

(Echchahed et al., 2025)

A Survey of State Representation Learning for Deep Reinforcement Learning

> *"An optimal representation can be defined by its ability to efficiently support policy learning for a set of downstream tasks. The learned space X should be constrained to a low dimensionality, while remaining sufficiently informative to enable the learning of an optimal policy (or value function) with limited-capacity function approximators. If X has too much information, it can slow down the learning process and hinder convergence to the optimal policy. Alternatively, a space with insufficient information will prevent convergence to the optimal policy (Abel, 2022). Hence X ideally balances well information capacity and simplicity."*

Comparing learned behavior to normative benchmarks under limited information, finite samples, or stale information

When a learner has limited information, finite data, or delayed updates, it should not be judged against an oracle that sees and knows everything. The right comparison is to a benchmark that matches the learner's information, update schedule, memory, and sample budget, or else to regret measures that make the gap to stronger benchmarks explicit. (1 source)

Once information constraints are taken seriously, the main question is not only "how far is learned behavior from the full-information optimum?" but also "which optimum is a fair target?" If the learner sees only partial state, has finite samples, or must act for long stretches without new observations or recomputation, then direct comparison to an oracle with full state and full model knowledge can mix together two very different gaps: one due to poor learning and one due to missing information. A normative comparison should separate these. In practice, this means comparing the learner first to the best policy feasible under the same observation and update limits, and only secondarily to a stronger oracle to show the price of those limits (Model-Generated).

This issue becomes especially clear when information is stale rather than absent. Musavi et al. study performance relative to "an oracle policy with full knowledge of the dynamics and continuous observations," but their regret analysis explicitly breaks the loss into parts caused by missed updates and by the intervals between updates. They show that, during those skipped intervals, the learner suffers not just from estimation error but from "model mismatch" as dynamics change and from "state and policy mismatch" because actions are based on outdated information and outdated computations. The resulting dynamic regret depends on temporal variation, uncertainty, and how long the agent goes without new information, with skip-interval costs growing with interval length unless mitigated by contraction in the dynamics (Musavi et al., 2026).

That decomposition suggests a good rule for evaluating learned behavior under limits. If the environment is changing, or if the learner updates only from time to time, then the right benchmark is often not a stationary full-information optimum at all, but a comparator that reflects what could be achieved with the same observation stream and same refresh rate. A stronger oracle can still be useful, but mainly as an upper bound or regret reference that measures the value of fresher information. Otherwise, one may wrongly call behavior "suboptimal" when much of the gap is simply the unavoidable cost of acting on old data (Musavi et al., 2026) (Model-Generated).

The same logic applies to finite-sample learning. With limited data, the learner cannot usually identify the true optimal policy, so a fair normative benchmark should account for statistical uncertainty and restricted coverage of the state-action space. In other words, we should ask whether the learned policy is good relative to what was learnable from the available samples, not only whether it matches the policy that would be chosen by an agent with full knowledge of the environment (Model-Generated). This is why sample-efficient

RL is often evaluated with regret, PAC guarantees, confidence sets, or offline-policy comparison under the same dataset, all of which try to measure performance against a benchmark that respects what the learner could realistically infer from its data (Model-Generated).

So the broad answer is: compare learned behavior to the benchmark implied by the learner's actual information structure. Use full-information optimal policies as ideal upper bounds, but use constrained optima, dynamic comparators, or sample-aware regret benchmarks to decide whether the behavior is normatively good given limited observation, finite data, or stale updates. That is the cleanest way to distinguish a failure to learn from the unavoidable cost of not knowing enough soon enough (Musavi et al., 2026) (Model-Generated).

Evidence

(Musavi et al., 2026)

Dynamic Regret in Time-varying MDPs with Intermittent Information

> *"We analyze its performance via dynamic regret relative to an oracle policy with full knowledge of the dynamics and continuous observations. Our main result establishes a dynamic regret bound that explicitly quantifies the impact of intermittent updates, decomposing regret into contributions from update times and skip intervals and revealing its dependence on temporal variation, estimation uncertainty, and the duration of intervals without updates. In particular, the dominant contribution from skip intervals admits a linear dependence on the interval length and the rate of temporal variation, while its effect is mitigated by mixing-induced contraction"*
>
> *". A central question in this setting is: How does performance degrade when the decision-maker is forced to operate for extended periods without new information or updates?"*
>
> *"During these intervals, two sources of error arise: (i) model mismatch, due to temporal variation in the dynamics, and (ii) state and policy mismatch, due to acting based on outdated information and outdated computations."*